COURSE • MY SUBJECT

# Session title

## A clear subtitle
## possibly on two lines

First Last • University • 2026

# Session outline

1. Boxes and callouts
2. Tables and code, with a longer line to show the automatic line wrapping
3. Diagrams, circuits and plots
4. Questions and wrap-up

**Goal**

One sentence that sums up what students will be able to do at the end.

# The colorbeam palette

**grape** #2A3F73
**midnight blue**
default, intro

**sky** #2F6DA3
**steel blue**
section, information

**mint** #1D7874
**teal**
section

**leaf** #4F8A3C
**green**
section, success

**sun** #C28A12
**amber**
tips, highlights

**clay** #B8622B
**terracotta**
section

**coral** #B5443B
**muted red**
warnings, questions

**rose** #A8416B
**raspberry**
section

**plum** #6A4A9C
**violet**
section

**slate** #55657A
**slate gray**
neutral, secondary

# Pills and numbered steps

## Steps in any color

1. Plan
2. Build
3. Test
4. Fix
5. Document

## Pills

grape sky mint leaf

sun clay coral rose

plum slate

An empty color, as in `\step{}{1}{5cm}`, uses the color of the current section.

# 1

# Part 1

## Boxes and callouts

highlight what matters

# Callout boxes



**card**

Neutral box, section color.

**warnbox**

A warning: do not miss this!

**tip**

A practical remark.

**okbox**

A result or a good practice.

**infobox**

Extra information or a reminder.

- Plain bullet list
- Bullets follow the section color

# Any color, any box



**colorcard{clay}**

Tinted background, colored frame.

**colorcard{rose}**

Use it for a recurring theme.

**colorcard{plum}**

Or to tell two ideas apart.

**colorcard{slate}**

Neutral side notes and references.

**colorcard{leaf}**

Same look as okbox, any color you like.

**darkbox: the one thing to remember**

# 2

# Part 2

## Tables and code

core, plus the programming library

# Comparison table

Part 2

| | Option A | Option B |
|---|---|---|
| **Criterion 1** | value | value |
| **Criterion 2** | value | **best value** |
| **Criterion 3** | value | value |
| **Criterion 4** | **worst value** | value |

# Code block (core)



example.py

```
def average(values):
    # compute the mean
    return sum(values) / len(values)

print(average([12, 15, 9]))
```

## Your turn

1. Write the code
2. Run it
3. Compare the result

# Python and JavaScript



**library: programming** `\usecolorbeamlibrary{programming}`

**mean.py**

```python
def mean(values):
    # arithmetic mean
    return sum(values) / len(values)

data = [12, 15, 9, 14]
print(mean(data))
```

**mean.js**

```javascript
const data = [12, 15, 9, 14];

function mean(v) {
  // arithmetic mean
  const s = v.reduce((a, b) => a+b);
  return s / v.length;
}

console.log(mean(data));
```

# C and C++

library: programming `\usecolorbeamlibrary{programming}`

**mean.c**

```c
#include <stdio.h>

int main(void) {
  int v[] = {12, 15, 9, 14};
  int s = 0;
  for (int i = 0; i < 4; i++)
    s += v[i];  /* sum */
  printf("%.1f\n", s / 4.0);
  return 0;
}
```

**mean.cpp**

```cpp
#include <iostream>
#include <vector>

int main() {
  std::vector<int> v{12, 15, 9, 14};
  double s = 0;
  for (int x : v) s += x;
  std::cout << s / v.size()
            << std::endl;
}
```

# Arduino, two blocks side by side



library: programming `\usecolorbeamlibrary{programming}`

**setup**

```
const int LED = 2;

void setup() {
  pinMode(LED, OUTPUT);
  Serial.begin(115200);
}
```

**loop**

```
void loop() {
  digitalWrite(LED, HIGH);  // on
  delay(500);
  digitalWrite(LED, LOW);   // off
  delay(500);
}
```

3

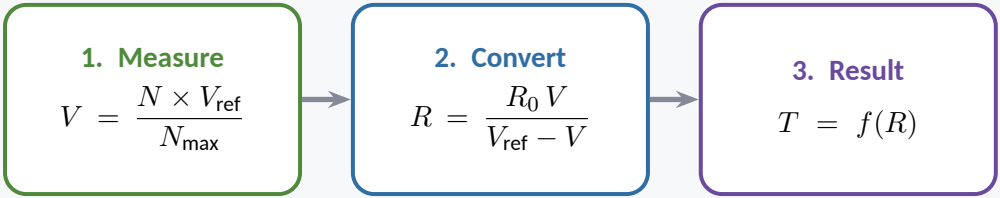
# Part 3

## Diagrams and plots

TikZ, pgfplots and the electronics library

# Chained formulas with flowbox





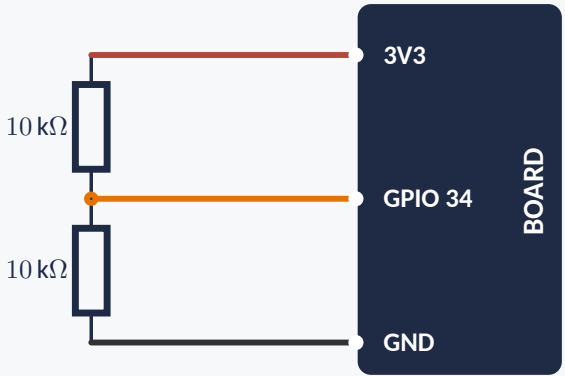

`flowbox` takes an optional color: `flowbox=plum`. Without it, it uses the section color.

# Electrical schematic



library: electrics

```
\usecolorbeamlibrary{electronics}
```



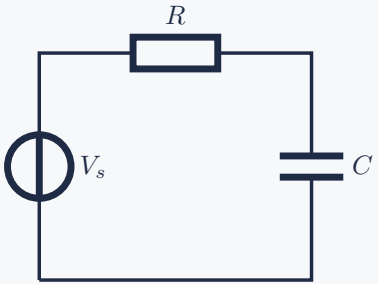

Voltage divider: the middle point is wired to the analog input.

Red: power. Black: ground. Orange: signal.

# RC charging circuit



**library: electronics** `\usecolorbeamlibrary{electronics}`



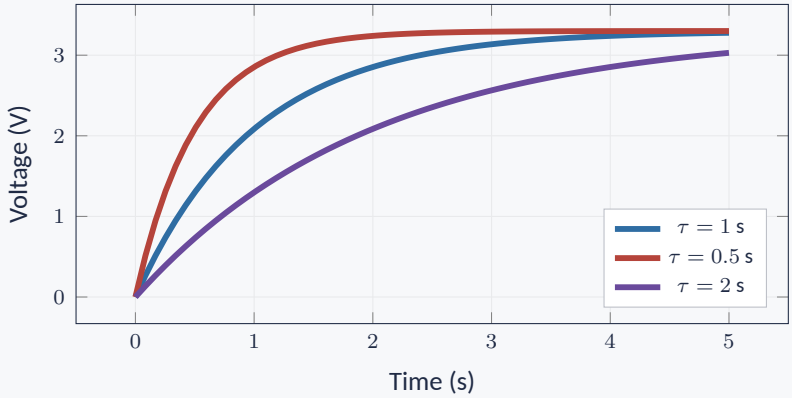

**Time constant**

$\tau = RC$, for example $10\,\text{k}\Omega \times 100\,\mu\text{F} = 1\,\text{s}$.

The next slide plots three values of $\tau$.

# Plot with several colors





# 4

# Part 4

## Questions and wrap-up

engage the class

# Question



**A question to ask the class?**

**Why?**

Discuss with your neighbor before moving on.



# Numbered rows and troubleshooting rows

Part 4

1 **Numbered row**

A title and a one-line description, framed in the section color

2 **Numbered row in another color**

Pass any palette color as first argument

| | |
|---|---|
| **Problem** | Short explanation of the cause and of the fix |
| **Another problem** | Same row, in red for something critical |

# Summary

1. First takeaway
2. Second takeaway
3. Third takeaway

**Going further: an idea for the next session**